

Gonçalo Soares

Senior Programmer / Stockholm, Sweden

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I am a gameplay and engine programmer with seven years of professional experience. I love games as an art form, and when I'm not making them I'm usually playing them.
I enjoy shaping features together with designers and artists while they are still ideas, iterating on them until we find something we can be proud of.
I also like making other developers faster by building tools and systems that other developers enjoy using.

EXPERIENCE

2023 – present
Stockholm, SE

Senior Programmer
Envar Games

- Used Unreal Engine 5 and AngelScript, building Witchspire from the start of pre-production through Early Access and on towards 1.0.
- Worked on multiple gameplay and core systems: abilities, enemy AI, networking, movement, inventory, crafting and UI, among others.
- The automation pipeline I built on top of Horde runs our development jobs, compiling changes and shipping them to Steam so we can get builds in front of playtesters quickly.
- Some of that work went back upstream, as fixes and extensions to Hazelight's open-source AngelScript fork of Unreal Engine.

2022 – 2023
Stockholm, SE

Mobile Engineer
King

- My area was In-App Purchases and Economy, working in C++ and TypeScript.
- I built new ways to present and select in-game offers.
- Designing and shipping the delivery tech meant working closely with producers, designers, data scientists and artists.
- Introducing some Agile practices from my time at Miniclip made our planning and sprints more predictable, and more enjoyable.

2019 – 2021
Lisbon, PT

Software Developer
Miniclip

- Shipped features across gameplay, AI, social, engine and tooling.
- Optimized our development tools. One of them was a Python script that validated our configuration data, a very common workflow — rewriting it in Rust took the validation from two minutes to one second.
- Improvements and fixes in the in-house engine, in C++, Metal, OpenGL and GLSL.

2019, 2020
Lisbon, PT

Guest Lecturer
Instituto Superior Técnico

- Invited back twice to lecture the Master's 3D Programming class on hardware ray tracing, and what it changes for real-time light transport.

EDUCATION

2017 – 2020
Lisbon, PT

Master's Degree in Information Systems and Computer Engineering
Instituto Superior Técnico

- Specialised in Game Development, Interactions and Visualization, with a thesis on real-time ray tracing with an AI-accelerated denoiser.

2014 – 2017
Lisbon, PT

Bachelor's Degree in Information Systems and Computer Engineering
Instituto Superior Técnico

OTHER PROJECTS

2025
Game jam

Fork Loop
GMTK Game Jam 2025 · itch.io

- Fork Loop is a speedrunning FPS where time rewinds between rounds and each round adds more enemies. The forks you throw remember their past lives and repeat them, so you spend most of it dodging your own daggers.
- Four of us made it over a jam weekend. It was a very fun way to explore a time rewinding mechanic.

2020
C++, Vulkan

Lift
Master's thesis · github.com/GoncaloFDS/Lift

- Lift is a real-time ray tracing renderer with an AI-accelerated denoiser, built with Vulkan and C++. Every frame is path traced and then denoised by NVIDIA's OptiX, which is what keeps it real-time instead of progressive.
- Path tracing and bidirectional path tracing both live in the same codebase, so I could compare which scenes each one struggles with.
- This work was presented at WSCG 2021 as *Lift: An Educational Interactive Stochastic Ray Tracing Framework with AI-Accelerated Denoiser*, with Professor João Madeiras Pereira.

SKILLS

Languages	C++ · AngelScript · Rust · Python · TypeScript · C · Lua
Gameplay Systems	Abilities · Movement · Inventory & crafting · Artificial Intelligence · Procedural generation · Networking
Engines & Tools	Unreal Engine 5 · Custom C++/Rust engines · Unity · Godot · Automation · Jenkins · Horde · Git · Perforce
Performance	Profiling · Optimization · Automation
Ways of Working	Code Review · Mentoring · Working across disciplines